Machine Learning Project: Predicting the Victories of Olympique Lyonnais

Walk-Forward Validation, Probabilistic Forecasts, Monte Carlo Season Simulation, and a Test Against the Betting Market

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Disclaimer. *I hope the ML does not predict well, because what the f***, OL will be 5th this season. (The simulation puts the average final position of OL at 4.9. So much for my hopes.) Joke aside, let us begin.*

Abstract

We ask whether machine learning can forecast the results of Olympique Lyonnais (OL) better than an Elo rating or the betting market. On 29,239 matches of the five major European leagues (2010/11 to October 2026) we build features strictly from information available before kick-off, checked by an automated perturbation test, and compare Elo, Poisson, logistic regression, random forest, gradient boosting, SVM and neural networks on a walk-forward test of 5,602 matches. (i) The *chosen model* is a multinomial logistic regression on 100 variables, refitted every season. (ii) Boosting and neural networks are statistically indistinguishable from it (log loss 0.979); all beat Elo (0.989) but none beats the bookmaker (0.966). (iii) Pooling our probabilities with the market's adds nothing: the optimal weight on the model is zero. (iv) Elo and squad market value explain almost everything that is predictable. A Monte Carlo simulator, backtested on 2023–2025, projects OL at 57.6 points (80% interval 49–66), an average final position of 4.9 and a 50% chance of the top four. A Champions League extension finds no clear favourite: five clubs sit around 13%.

1 Introduction

Football is the sport where the gap between a simple forecast and the best one is smallest: a match is decided by a handful of goals, the stronger team wins only about one time in two, and one game in four ends in a draw. This makes it a good laboratory for a question that also matters in finance: *how much can a model extract from public data, how reliably can that be measured, and does it survive comparison with a liquid market?* OL plays about 35 league matches a year, far too few to train a model. I therefore train on all five major leagues (Premier League, La Liga, Serie A, Bundesliga, Ligue 1) and use OL as a case study. The object to forecast is the following.

Definition 1.1 (Match forecast)**.** A *match forecast* is a probability vector $p = (p_A, p_D, p_H)$ over the outcomes away win, draw and home win, computed from information available before kick-off. The *expected points* of the home side are $3p_H + p_D$.

For instance, for Paris SG–Lyon on 26 October 2026, the model gives the forecast of Table 1, close to the bookmaker's.

Table 1: Forecast for Paris SG–Lyon, 26 October 2026.

	OL win	Draw	OL loss	OL expected points
Model	15%	21%	64%	0.67
Bookmaker (margin removed)	15%	20%	65%	–

The model I chose. A *multinomial logistic regression* on 100 input variables (Elo rating, squad market value, recent form, shots, schedule), refitted once a year (Section 4). I also tested random forest, gradient boosting, a support-vector machine and neural networks, all with scikit-learn. None is significantly better, so I kept the simplest and most explainable one. **Three rules.** (1) *No leakage*: a variable used to predict a match played at date T contains nothing dated T or later. (2) *Temporal validation*: train on the past, judge on the future, never on a random split. (3) *Honest evidence*: every claimed improvement comes with a confidence interval, and negative results are reported as negative.

2 Data

Table 2: Data sources.

Source	Content	Use
football-data.co.uk	Results, shots, corners, cards, bookmaker odds; 5 leagues, 2010/11–2026	Core data
Transfermarkt dataset (CC0)	656,000 player market valuations	Squad value
Self-computed	Elo ratings, shot-based "xG proxy"	Variables
Wikipedia	Champions League 2026/27 fixtures and scores	Section 13

The sample has 29,239 matches (593 for OL) and 189 clubs: training 2011–2020 (about 18,200 matches), validation 2021–22 (3,651) and test October 2023 to October 2026 (5,602). **Rejected sources.** FBref advanced statistics were removed by their provider in January 2026; Understat forbids automated access in its `robots.txt`; ClubElo was unreachable. No free historical source exists for injuries or line-ups, so they could not be tested. **Terms of use.** football-data.co.uk allows private, non-commercial use only. This is a private academic study: raw data are downloaded, kept out of the repository and not redistributed.

3 Variables and leakage control

Every variable is computed for the home team, the away team and their difference, from matches strictly before the one predicted.

Definition 3.1 (Leakage-free variable)**.** A variable attached to a match played at date T is *leakage-free* if it is a function of results dated strictly before T only.

Table 3: Families of variables.

Family	Construction
Elo rating	Strength score updated after every match (below), one scale per league
Recent form	Mean over the last 3, 5, 10 matches of points, goals for and against, shots on target
Venue form	Last 5 home (or away) matches
League table	Points and goal difference per match, position before the match
Schedule	Rest days, matches in the previous 14 days
Squad value	$\ln V$ of the team minus the league average of that month (Transfermarkt)

Elo [4]**.** Before a match between i (home) and j (away), the expected score of the home team is

$$E_{ij} = \frac{1}{1 + 10^{-(R_i - R_j + 65)/400}}, \tag{1}$$

where 65 points is the home advantage. With $S = 1, \frac{1}{2}, 0$ for a win, draw, loss and g the goal difference, the ratings are updated after the match by

$$R_i' = R_i + K\,(1 + \ln(1 + |g|))\,(S - E_{ij}), \qquad R_j' = R_j - (R_i' - R_i), \qquad K = 20. \tag{2}$$

The rating stored for the match is the one *before* this update. **Squad value** is measured relative to the league so that market inflation is not mistaken for strength:

$$sv_i = \ln V_i - \frac{1}{|L|} \sum_{k \in L} \ln V_k. \tag{3}$$

How leakage is prevented. Rolling variables are shifted by one match; Elo is stored before its update; valuations used for the squad value are dated before the monthly snapshot; bookmaker odds are never a variable. An automated test replaces every result after a cutoff date by random values, reruns the whole pipeline and requires all earlier variables to be identical. It fails, as it should, on two bugs I injected on purpose (a rolling window without the shift, Elo stored after its update). **Two lessons from the data.** The "xG proxy" (a regression of goals on shots) collapses to 0.287× shots on target, so it is not a real xG. Variables counting how many matches a team has been observed for are just a calendar clock; I removed them (cost +0.0011 log loss on validation, not significant).

4 The model I chose: multinomial logistic regression

Definition 4.1 (Multinomial logistic regression)**.** Let $x \in \mathbb{R}^{100}$ be the (standardised) variables of a match. For each outcome $k \in \{A, D, H\}$ the model computes a linear score $z_k = w_k^\top x + b_k$ and returns

$$P(\text{outcome} = k \mid x) = \frac{\exp(w_k^\top x + b_k)}{\sum_{j \in \{A,D,H\}} \exp(w_j^\top x + b_j)}. \tag{4}$$

Estimation. The weights maximise the log-likelihood of the training matches, with an L^2 penalty that keeps them small:

$$\min_{W,b} \; -\frac{1}{N} \sum_{i=1}^{N} \ln P(y_i \mid x_i) \; + \; \frac{1}{2CN} \, \|W\|^2. \tag{5}$$

The strength $1/C$ of the penalty is chosen by *time-ordered* cross-validation inside the training seasons (each fold trains on the past and scores the next block). Variables are standardised and missing values replaced by the median, both fitted on training data only. The model is refitted once a year.

Remark 4.1 (Why this model). (a) It is as accurate as the alternatives (Section 7). (b) It is explainable: the linear scores split exactly into the contribution of each family of variables, $z_k = b_k + \sum_f c_{k,f}$, which the website displays. (c) Its probabilities are well calibrated. (d) It is fast enough to simulate 10,000 seasons per league.

Table 4: Models compared.

Family	**Models compared**
Baselines	Class frequencies; league table; Elo calibrated to 3 classes; independent Poisson goals [5]
Linear (chosen)	Multinomial logistic regression with L^2 penalty
Trees	Random forest; gradient boosting
Kernel	RBF support-vector machine (under-tuned)
Neural	MLP with 1 and 2–3 hidden layers (L^2, few epochs, mean of 5 seeds)
Online	Logistic regression whose weights are updated by gradient descent after each week of results
Reference	Bet365 prices turned into probabilities (margin removed, average margin 5.5%)

Only scikit-learn [3] and numpy are used. Dropout and Batch Normalisation do not exist in scikit-learn; I replaced them by an L^2 penalty, few epochs and seed averaging. LSTM and GRU networks were not built: the rolling variables already summarise each team's history.

5 How quality is measured

Accuracy is a weak summary here: a draw is almost never the most likely outcome, so a useful model can rarely name one. The main metric is the log loss, a strictly proper scoring rule [2].

Definition 5.1 (Log loss)**.** For N matches, with p_i the probability given to the outcome that actually occurred in match i,

$$\mathrm{LL} = -\frac{1}{N}\sum_{i=1}^{N} \ln p_i. \tag{6}$$

Example. If the model gave 20% to what happened, the match costs $\ln 5 = 1.61$; if it gave 60%, it costs 0.51. A model that always says $(30.0\%, 25.2\%, 44.7\%)$ (the average frequencies) scores 1.075; a model that knew the result would score 0. I also report the Brier score [1] and the expected calibration error:

$$\mathrm{Brier} = \frac{1}{N}\sum_{i=1}^{N}\sum_{k}(p_{ik} - o_{ik})^2, \qquad \mathrm{ECE} = \sum_{b}\frac{n_b}{N}\,|\bar{p}_b - \bar{o}_b|, \tag{7}$$

where $o_{ik} = 1$ if outcome k occurred in match i and 0 otherwise, and b indexes bins of forecast probabilities. **Significance.** To compare models A and B, I take the paired difference $\Delta = \overline{\ell_A - \ell_B}$ of their per-match log losses and build a 95% confidence interval by a *block bootstrap* that resamples whole calendar weeks (matches of the same week are not independent), with 4,000 draws. With 5,600 test matches, a difference below about 0.003 cannot be detected.

6 Validation design

Football results form a time series: a random split would let the model learn from matches played after those it is tested on. Training covers 2011–2020, validation 2021–22 (used to compare models during development) and test October 2023 to October 2026 (used once, with frozen hyper-parameters). The final evaluation is *walk-forward*: each test season S is predicted by models fitted on seasons $2011, \ldots, S-1$, as it would run in real life.

7 Results

Table 5: Walk-forward test, October 2023 to October 2026 (5,602 matches).

Model	Accuracy	Log loss [95% CI]	Brier	ECE
Baseline (frequencies)	43.2%	1.0754 [1.069, 1.082]	0.6509	0.012
Elo	52.5%	0.9888 [0.979, 0.999]	0.5900	0.017
Poisson goals	52.0%	0.9902 [0.980, 1.001]	0.5905	0.022
Logistic regression (chosen)	**53.3%**	**0.9793 [0.969, 0.990]**	**0.5834**	**0.016**
Gradient boosting	53.0%	0.9787 [0.968, 0.989]	0.5826	0.020
Neural network (MLP)	53.0%	0.9796 [0.969, 0.990]	0.5834	0.021
Ensemble of the three	53.0%	0.9779 [0.967, 0.988]	0.5824	0.017
Bookmaker (Bet365)	53.9%	0.9658 [0.955, 0.977]	0.5740	0.022

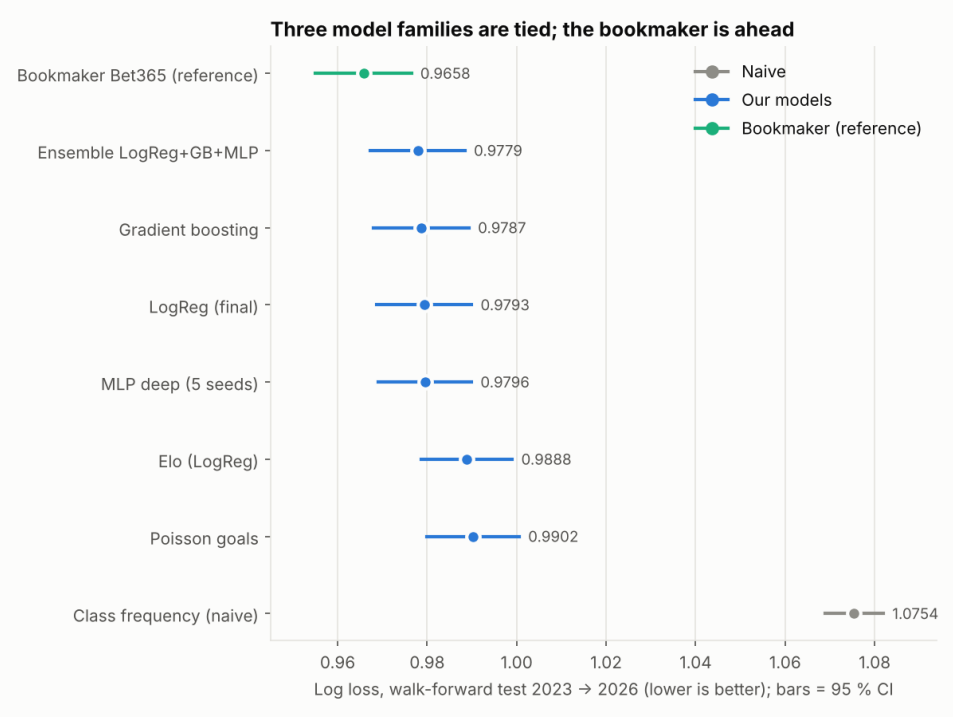


Figure 1: Log loss on the test seasons with 95% confidence intervals: three model families are tied, the bookmaker is ahead.

Result 7.1. The logistic regression beats Elo ($\Delta = -0.0095$, 95% CI $[-0.0135, -0.0056]$). Gradient boosting (-0.0007) and the neural network ($+0.0003$) do *not* beat it: their intervals

contain 0. The bookmaker beats every model ($\Delta = +0.0135$ for the logistic regression, CI $[+0.0094, +0.0175]$). Complexity did not pay.

7.1 Calibration

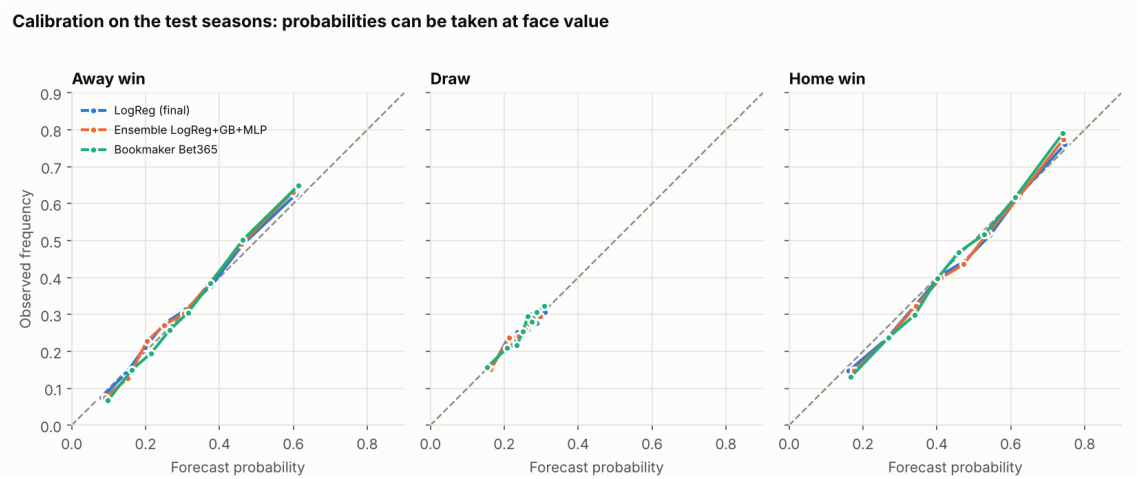


Figure 2: Reliability diagrams on the test seasons: the closer to the diagonal, the more honest the forecast.

The probabilities are well calibrated (ECE between 1.6% and 2.2%). The model almost never names a draw as the most likely outcome (6 matches out of 5,602): draws are essentially unpredictable.

7.2 Ablation: which data add value?

Table 6: Ablation: nested blocks of variables (validation seasons).

Logistic regression, validation	Log loss	Change vs previous [95% CI]
A: Elo only	0.9957	–
B: + results form	0.9931	-0.0026 $[-0.0051, -0.0001]$
C: + shots and xG proxy	0.9887	-0.0044 $[-0.0072, -0.0017]$
D: + league table, fatigue	0.9890	$+0.0003$ $[-0.0013, +0.0019]$ (nothing)
E: + squad value (final)	0.9870	-0.0020 $[-0.0048, +0.0009]$

Form and shots each add a small but detectable gain on top of Elo; league position and fatigue add nothing. Squad value is not significant on two validation seasons although it ranks second in the importance analysis below: it overlaps with Elo, both measuring long-run strength.

8 What drives the predictions?

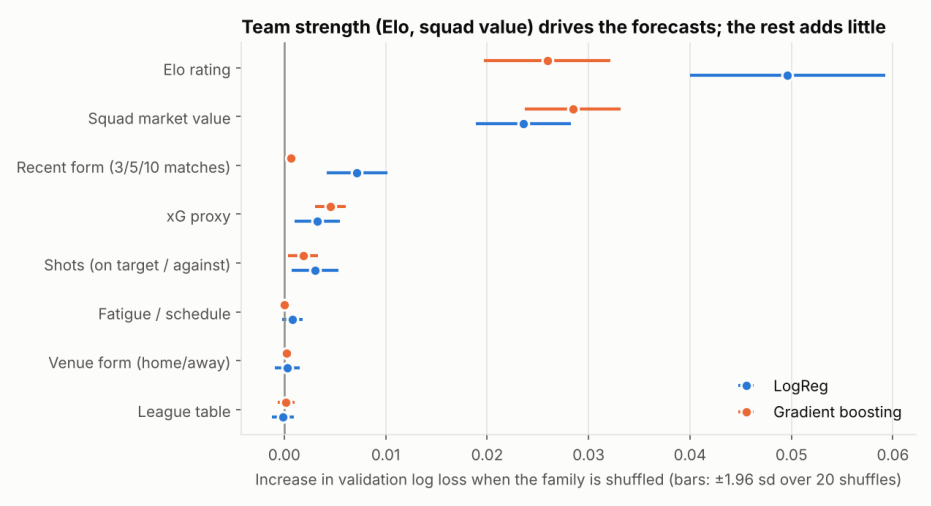


Figure 3: Increase of the validation log loss when a family of variables is shuffled (permutation importance).

Elo (+0.050) and squad value (+0.024) dominate. Recent form adds +0.007 and shots +0.003; rest days, league table and venue form are close to zero. Home sides win 44.7% of matches, draw 25.2% and lose 30.0%, which sits in the intercept.

Remark 8.1. In the linear model, form has a slightly negative effect at fixed Elo and squad value. This is a collinearity artefact (overlapping windows), not evidence that a good run hurts. Importance describes what the *model* uses, not what causes results.

9 Model versus bookmaker

This is an academic question only; no betting strategy is built. To test whether the model contains information the market lacks, I pool the two by log-linear pooling,

$$p \;\propto\; p_{\text{market}}^{\,1-w}\, p_{\text{model}}^{\,w}, \tag{8}$$

and choose the weight w on validation. On the test seasons the best weight on the model is $w = 0.00$: the market already contains everything the model knows. The pooled log loss (0.9663) is not better than the market alone (0.9658). The market takes a margin of 5.5% on average and is still sharper than the model; the likely reason is information I do not have (line-ups, injuries, news).

10 Previous results: was the model right?

Remark 10.1. These forecasts are recomputed *after the fact* with the walk-forward method (models fitted only on earlier seasons, variables built only from earlier matches). They were **not published in advance**. "Right" means that the most likely outcome occurred: a 45% pick is wrong more often than right, by design, so log loss and calibration are the fair scores.

Table 7: Accuracy and log loss (LL) over recent periods.

Period	Matches	Correct	LL model	LL bookmaker	LL no-skill
Last 7 days	49	57%	0.927	0.902	1.052
Last 30 days	96	56%	0.928	0.906	1.042
Whole test period (2023 → Oct 2026)	5602	53%	0.979	0.966	1.075

Table 8: When the model said x%, how often did it happen? (all 5,602 test matches)

Top pick given	Matches	Mean forecast	Came true
0–40%	802	38%	39%
40–50%	1791	45%	45%
50–60%	1440	55%	55%
60–70%	956	65%	63%
70–100%	613	76%	78%

When the top pick was given 60–70%, it happened 63% of the time. Over the last 30 days the model is right 56% of the time and is slightly less sharp than the bookmaker.

Table 9: The 14 most recent matches (probabilities in %).

Date	Match	Score	H / D / A	Top pick	Result
10-21	Levante – Ath Bilbao	3–1	34 / 28 / 38	Away win 38%	Wrong
10-19	Getafe – Vallecano	1–1	35 / 28 / 37	Away win 37%	Wrong
10-19	Parma – Torino	0–3	40 / 29 / 31	Home win 40%	Wrong
10-19	Monza – Cagliari	4–0	29 / 32 / 39	Away win 39%	Wrong
10-19	Tottenham – Coventry	3–1	50 / 30 / 20	Home win 50%	Right
10-18	Real Madrid – Sevilla	4–1	74 / 16 / 10	Home win 74%	Right
10-18	Osasuna – Santander	2–0	56 / 26 / 19	Home win 56%	Right
10-18	Malaga – Sociedad	2–5	26 / 27 / 48	Away win 48%	Right
10-18	Celta – Alaves	2–1	52 / 25 / 23	Home win 52%	Right
10-18	Udinese – Lecce	0–3	58 / 24 / 18	Home win 58%	Wrong
10-18	Milan – Atalanta	3–2	52 / 25 / 24	Home win 52%	Right
10-18	Juventus – Lazio	0–3	62 / 22 / 16	Home win 62%	Wrong
10-18	Fiorentina – Como	1–3	22 / 27 / 51	Away win 51%	Right
10-18	Paris FC – Rennes	4–3	45 / 27 / 28	Home win 45%	Right

11 Case study: Olympique Lyonnais

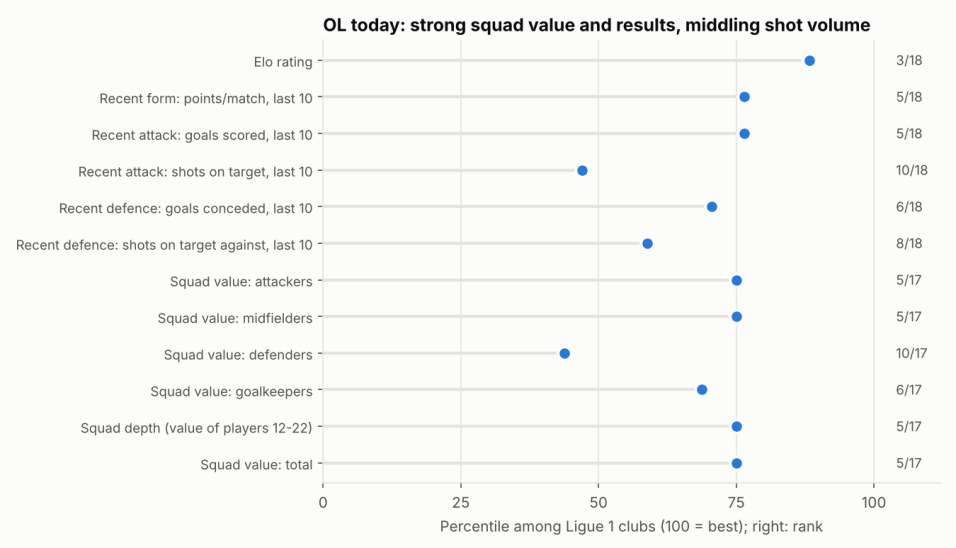


Figure 4: OL's profile in Ligue 1 on 26 October 2026 (percentile among clubs; right: rank).

OL is third in Ligue 1 by Elo (1,615; Paris SG 1,707), fifth for recent points and goals scored, and fifth by squad value (EUR 305 m; the defence is only tenth). Results and rating are better than the shot volume (tenth for shots on target) would suggest. The 90% interval on OL's win probability at Paris SG is 14–17% with model uncertainty only (100 bootstrap refits) and 12–21% once uncertainty on the true strength of both teams is added (Elo shaken by 60 points).

12 Season simulation

Definition 12.1 (Simulated season)**.** Given the current state, a *simulated season* (i) shakes the Elo rating of every team by $\varepsilon \sim \mathcal{N}(0, 60^2)$, (ii) draws the outcome of each remaining match from the model's probabilities, (iii) draws a scoreline from a Poisson goal model consistent with that outcome (for goal-difference tie-breaks) and (iv) ranks the table on points, goal difference and goals scored. The expected points of a team are its current points plus $\sum_m (3p_{\text{win}}^{(m)} + p_{\text{draw}}^{(m)})$ over its remaining matches.

The shock size, 60 Elo points, was chosen by a *backtest*: each past season was simulated from 31 October using only what was known then, and compared with the real final table.

Table 10: Backtest on 2023–2025 (15 league-seasons, 288 team-seasons).

	Simulator	Naive reference
Mean absolute error of final points	6.4	8.9
Brier score, title	0.022	0.049
Brier score, top 4	0.064	0.165
Brier score, relegation	0.055	0.117
Coverage of the 80% interval	72%	80%

The simulator beats the naive reference, but its "80%" intervals cover only 72%: they should be

read as lower bounds.

Table 11: Projected top three of each league (by expected final position, 10,000 simulated seasons).

League	1st	2nd	3rd
Premier League	Man City (80 pts, title 57%)	Arsenal (76 pts, title 28%)	Brighton (71 pts, title 12%)
La Liga	Barcelona (98 pts, title 93%)	Real Madrid (84 pts, title 6%)	Ath Madrid (79 pts, title 2%)
Serie A	Inter (86 pts, title 71%)	Roma (78 pts, title 19%)	Como (72 pts, title 5%)
Bundesliga	Bayern Munich (83 pts, title 96%)	Dortmund (66 pts, title 3%)	Leverkusen (62 pts, title 1%)
Ligue 1	Paris SG (72 pts, title 77%)	Monaco (60 pts, title 7%)	Lille (59 pts, title 6%)

Table 12: Ligue 1 2026/27 (state of 26 October 2026, 7 matches played).

Team	Exp. points [80%]	Title	Champions Lg.	Top 4	Relegation
Paris SG	71.9 [64–80]	76.9%	96.2%	98.5%	0.0%
Monaco	59.9 [51–69]	7.2%	47.7%	62.3%	0.0%
Lille	59.0 [50–68]	5.7%	41.9%	56.6%	0.0%
Lyon	**57.6 [49–66]**	**4.0%**	**35.7%**	**50.4%**	**0.0%**

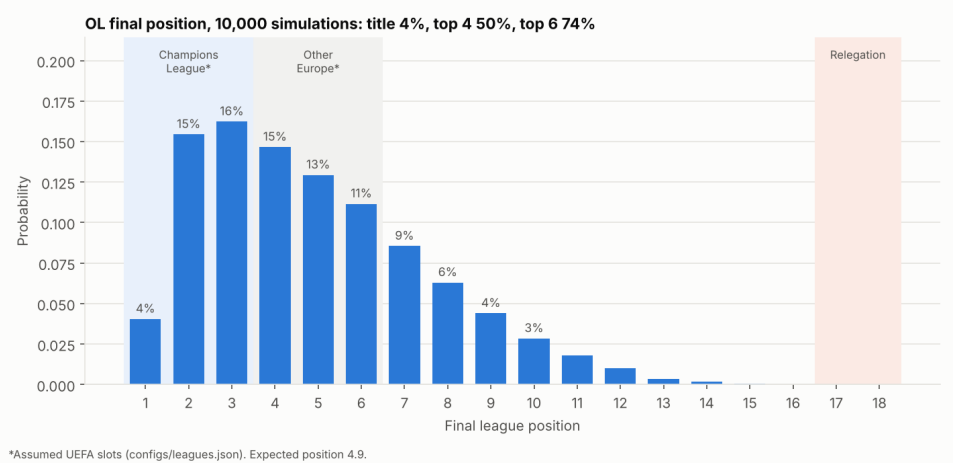


Figure 5: Final position of OL over 10,000 simulated seasons.

The average final position of OL is **4.9**; the most likely finishes are 2nd and 3rd (15% and 16%), but the distribution is wide: after 7 matches six clubs share 13–14 points, so the forecast rests on team strength rather than on the table.

13 Champions League 2026/27

The aim is to use the real fixtures and to combine each club's performance in its own championship. The calendar (144 league-phase matches, 36 clubs) comes from Wikipedia; only 29 matches were recorded when I collected it, the others are simulated. Domestic leagues are only comparable through matches between their clubs, so I fitted a logistic regression on 3,085 past Champions League, Europa League and Conference League matches (2012–2025):

$$d^{\text{value}} = \ln V_{\text{home}} - \ln V_{\text{away}}, \qquad d^{\text{ppg}} = \text{ppg}_{\text{home}} - \text{ppg}_{\text{away}}, \tag{9}$$

where ppg is the number of domestic league points per match over the last 20 league matches.

Table 13: European strength model.

Walk-forward 2023–2025 (943 matches)	**Log loss**
Class frequency	1.0351
Squad value only	0.9840
Domestic form only	1.0368
Value + domestic form	0.9834

Result 13.1. Squad value carries the signal (-0.051 vs the class frequencies, significant). Domestic form adds nothing measurable once value is known (-0.0005, 95% CI $[-0.0048, +0.0037]$). I keep it because the question asked to combine both, but its weight is small.

I then simulated 20,000 times the rest of the league phase, the play-offs (ranks 9–24), the round of 16, quarter-finals, semi-finals and the final (two legs; extra time and penalties close to a coin flip; the final on neutral ground).

Table 14: Champions League 2026/27: probabilities from 20,000 simulations.

Club	**Top 8**	**Semi-final**	**Final**	**Winner**
Arsenal	73%	40%	23%	13.5%
Manchester City	47%	37%	23%	13.4%
Barcelona	60%	38%	23%	13.3%
Real Madrid	86%	39%	23%	12.7%
Paris Saint-Germain	76%	39%	23%	12.6%
Bayern Munich	67%	34%	19%	9.8%
Liverpool	70%	27%	13%	6.1%
Inter Milan	27%	16%	7%	3.0%
Manchester United	18%	14%	6%	2.7%
Sporting CP	26%	12%	5%	1.8%

There is **no clear favourite**: Arsenal, Manchester City, Barcelona, Real Madrid and Paris SG are each around 13%, Bayern 10%, Liverpool 6%; the differences between the first five are smaller than the uncertainty. This model is validated match by match, not on whole tournaments; the knockout bracket is approximate; AEK Athens, Sabah and Slovan Bratislava are not in the data and are assumed weak. No Europa League winner is predicted yet.

14 Limitations

- No injuries, suspensions, line-ups or coaching news (the likely reason the bookmaker is better); no real xG.
- Squad values are a crowd-sourced snapshot (June 2026); 13 clubs have no value.
- The bookmaker benchmark uses Bet365 pre-match prices with a simple margin removal.
- Few validation windows; the SVM is under-tuned; online updating was tested for the linear model only.

- Simulation: intervals are too narrow in the backtest, UEFA places are assumptions, and strength is constant apart from a random shock.
- Team state excludes the results of 23–25 October 2026, not yet in the data source.

15 Conclusion

On public data, football results can be forecast with *well-calibrated probabilities that are only modestly sharper than team strength alone*. Elo and squad value explain most of what is predictable. The chosen model, a regularised multinomial logistic regression, is as good as boosting and neural networks: complexity did not pay. The models cannot beat the betting market and add no information to it, which is consistent with an efficient market. For OL, the model sees a top-five squad with results slightly ahead of its shot volume: about 57.6 points, an average finish of 4.9 and a coin-flip chance of the top four. So the disclaimer at the beginning stands: I really hope the ML is wrong. **Reproducibility.** Code, tests (including the leakage test) and the interactive website are in the repository; `make pipeline` rebuilds data, variables, OL predictions and simulations. All numbers come from `reports/tables/`.

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